

Errors reveal the depth of human reasoning

Measuring depth of deliberation from the structure of mistakes
in millions of real decisions

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Author note. *Results report held-out analyses on the full dataset; strata and the primary interaction were fixed before held-out evaluation. Evaluation is split by player throughout.*

Abstract

How deeply a person deliberates before deciding is central to theories of bounded rationality, yet it is rarely measurable in natural behaviour. Here we show that the depth of human reasoning can be read directly from the structure of the mistakes people make. We use competitive chess as a natural laboratory in which millions of real, skill-graded decisions are scored against an authoritative standard at progressively deeper levels of analysis. Players of different skill differ little on the moves they get right but are sharply separated by the depth at which their errors are exposed: stronger players are misled only by deeper traps, a quantity we term the *depth of satisficing*. Within each time control this depth rises with skill, and it collapses under time pressure—vanishing entirely in ultra-fast play, a built-in control. Fit without any timing information, it nonetheless tracks how long players actually think, and on synthetic players who stop at known depths it recovers the planted depth. A predictive model incorporating inferred depth *selectively* outperforms a state-of-the-art behavioural model on slow, high-swing decisions—where deep reasoning can vary—while adding nothing where decisions are fast or shallow. Estimated per individual, the construct yields an interpretable cognitive profile that separates elite players a one-dimensional score misranks. These results indicate that people commit to the best option found at the depth of search they reach, and that what someone gets wrong, more than what they get right, reveals how they think.

1 Introduction

A long tradition in the decision sciences holds that people do not optimise but *satisfice*: they search through options until one crosses an acceptability threshold, then stop and choose (Simon, 1956). Allied ideas—bounded rationality, level- k reasoning, and the contrast between fast intuitive and slow deliberative thought (Kahneman and Tversky, 1979; Stahl and Wilson, 1995)—all turn on a hidden quantity: how far a decision maker looks before committing. That quantity has been inferred almost exclusively from small staged experiments in artificial settings. Whether the regularities found there govern real, repeated, high-stakes choice is hard to test, because natural decisions rarely come with a ground-truth measure of how good each option was, let alone how good it *looked* at successive depths of consideration.

Chess supplies exactly this. Millions of decisions by players of precisely known skill are recorded in real competition, and each can be scored against an engine far stronger than any human—at every depth of an iterative-deepening search. The engine’s evaluation of a move *changes with depth*: a move that looks strong shallow may be refuted only deeper, and a move that looks weak at first may reveal its worth later. This depth-resolved record lets us ask, of every human mistake, *at what depth the flaw becomes visible*, and therefore how deep the player must have looked.

Our central construct is the **depth of satisficing**: the depth at which the move a player chose stops looking better than the move they should have chosen. We show that this is the dimension along which skill is most legible. Players of widely different ratings are nearly indistinguishable on decisions they get right; the information separating them is concentrated in the small fraction of moves on which they err, and specifically in the depth at which those errors surface.

We make three advances over prior chess-error and engine-depth work (Regan and Haworth, 2011; Biswas and Regan, 2015; McIlroy-Young et al., 2020). First, we replace descriptive curve-fitting with a generative model in which the player is a latent distribution over search depths, so depth of satisficing is a posterior quantity with stated uncertainty. Second, we validate that engine depth is a meaningful proxy for human reasoning depth in three independent ways—predictive, convergent, and differential—rather than assuming it. Third, we use inferred depth to build interpretable individual profiles and to predict held-out human moves, outperforming a strong state-only behavioural model precisely where theory says deep reasoning should matter, but not elsewhere; this is notable given that strong searchless models exist (Ruoss et al., 2024).

2 Results

Skill lives in the errors, not the successes. Across 922,412 decisions spanning ratings 1200–2600+ and all time controls, the error of the played move is nearly indistinguishable across skill at shallow depth and diverges only as depth increases: the mean regret of the played move at full depth falls from 0.063 at the weakest band to 0.029 at the strongest (Fig. 1). The discriminating information is concentrated in the depth structure of mistakes: a single “swing-down” decision—a played move that looks good shallow but is overtaken deeper—carries $\approx 6\times$ the rating information of an average decision ($R^2 = 0.081$ vs. 0.012 for predicting rating from one move), and that information lies in *at what depth* the move is refuted, not in the size of the error.

Depth of satisficing rises with skill and collapses under time pressure. Because stronger players are over-represented in slow play, this relationship must be read *within* a time control; pooling conflates skill with available clock. Within classical games the posterior depth of satisficing rises from $d \approx 10.8$ at the lowest band to $d \approx 13.7$ at the highest, with non-overlapping 95% credible intervals between the extreme bands (Spearman $\rho = +0.43$; rapid +0.53, blitz +0.43; Fig. 2a). In *bullet* the relationship vanishes ($\rho = -0.03$, flat across bands)—a built-in negative control: when no one has time to search, depth no longer separates skill, ruling out a pure rating artefact. The same logic appears within players: de-measured depth rises across think-time quartiles ($-0.11, -0.10, -0.07, +0.27$ plies, fast to slow; Fig. 2b).

A within-game time-pressure signature at the move-40 control. Over-the-board games impose a classical time control at move 40, after which the clock is replenished. In these games the error of the played move (in live, undecided positions) rises through the move 31–40 scramble and drops sharply once time is restored (Fig. 7), reproducing in win-probability terms a centipawn-based signature first reported by Biswas and Regan (2015). The effect is specific to games that carry the move-40 control: online “classical” play, which uses a single incremental control, shows no such peak, and averaging over decided endgames (where win probability is insensitive to error) erases it—so the signature emerges only when both conditions hold.

Inferred depth tracks real thinking time. This is the construct’s key non-circular test. A model fit with the clock feature removed—so timing can play no role in the estimate—nonetheless

yields a depth that correlates with the actual time the player spent on the move (held-out Spearman $\rho = +0.39$; opening +0.35, middlegame +0.44, endgame +0.30; Fig. 3), strongest in the middlegame where deliberation varies most. Inferred depth tracks real thinking time without ever being told about time—convergent evidence that it reflects deliberation rather than the rating prior.

A depth-aware model predicts human moves better, only where depth should matter.

A fusion of a state-only model (Maia-3) and a search-only mixture-over-depths model improves held-out cross-entropy over Maia-3 alone by a modest 0.012 nats overall (95% CI [0.008, 0.016]; top-1 match +0.7 points). The interest is not the average but the *differential*, registered in advance: the gain is concentrated in classical, high-swing positions (+0.053 nats) and is absent—even slightly negative—in fast or low-swing play (Fig. 4). The swing-magnitude $\times$ time-control interaction is +0.059 nats (95% CI [0.054, 0.063]; player-clustered bootstrap), confirmed by a mixed-effects model with players as random effects ($\beta = 0.057$ nats, $p < 10^{-200}$). A state-only model cannot produce this pattern. The gain further localises by swing *direction*: within classical play it falls almost entirely on swing-up moves (+0.048 nats)—those whose worth appears only at depth—so the depth signal predicts precisely the choices a pattern model cannot.

Individual profiles separate expert styles. Estimated per player from an *elo-free* model (so recovering rating is not circular), a four-component profile—depth of satisficing, trap-susceptibility, deep-discovery rate, and time-elasticity—recovers rating across 950 players (≥ 100 decisions each) by nested cross-validation (held-out $R^2 = 0.57$), whereas the depth axis *alone* has no predictive power ($R^2 \approx 0$). The profile resolves a known difficulty: among elite players a one-dimensional depth score is discordant with true rating on 10 of 15 pairs, because it would rank, e.g., Abdusattorov and Nepomniachtchi above Carlsen; the multi-axis profile instead attributes their strength to different axes—Nakamura is most trap-resistant yet least time-elastic (a blitz specialist who plays deep *fast*), Carlsen most converts thinking time into depth—and no longer inverts their ratings (Fig. 5).

The instrument recovers known depth. On synthetic agents that satisfice at fixed, known depths, recovered depth increases monotonically with the planted depth (group-level Spearman = 1.0 across planted depths 4–16; the estimator also independently recovers the agents’ decision sharpness, $\hat{\beta} = 7$ vs. generating $\beta = 6$; Fig. 6). The recovery is ordinal rather than exact—per-agent estimates are compressed toward the middle and noisy (mean absolute error 3.7 plies) given ~ 70 decisions per agent—but it confirms the model measures depth rather than position difficulty. Notably, recovery requires estimating depth from the moves under a flat prior; imposing the human rating-conditioned prior collapses the estimate toward its mean, which is why the rating relationship (Fig. 2) must be corroborated by the non-circular tests above.

3 Discussion

The errors people make are usually treated as noise. We find the opposite: where success is widely shared, error carries the informative variation, and its fine structure—the depth at which a flaw becomes visible—is a measurable signature of how a person reasons. Depth of satisficing scales with expertise, contracts under time pressure, and powers prediction exactly in the slow regime where depth can vary. This supports the view that people commit to the best option found at the plateau of search they reach rather than integrating across the whole trajectory, consistent with level- k choices being largely independent of higher-order belief tails (Strzalecki, 2014). In principle the

approach extends to any domain with skill-graded decisions and a depth-resolved value standard—a depth-resolved value network in Go, or staged value estimates in other tasks—which we leave as the natural next test of generality. A constructive corollary: an agent that estimates a person’s depth of satisficing could tailor problems just beyond it. *Limitations:* engine search depth is a *validated proxy* for human selective search—supported here by convergence with thinking time (Fig. 3) and by recovery of planted depth (Fig. 6), but not isomorphic to it; the depth-of-satisficing posterior leans partly on the rating prior, which is why we rely on those two non-circular tests; the aggregate predictive gain is small and the claim rests on its registered differential; and chess is adversarial and highly trained, so cross-domain generality remains to be demonstrated.

4 Methods

Data. We use 922,412 decisions (16,328 players): online games from the Lichess open database (official parquet export, month 2025-09), balanced-sampled across rating bands (200-point bins, 1200–2600+) and time classes (bullet/blitz = fast, rapid/classical = slow), with per-move clock times retained; and an elite over-the-board stratum of 371,279 decisions from 1,261 titled players (broadcast tournament games), which supports the individual-profile case study. Synthetic agents (below) provide ground truth. All decisions are filtered to plies 9–120 to skip opening-book play.

Engine analysis and value scaling. Positions are analysed with Stockfish 18 in Multi-PV mode, recording each candidate move m_i ’s evaluation $v_{i,d}$ at every depth $d = 1 \dots D$ ($D=21$, $K=9$). The candidate set is the union of top- K across all depths so trap moves keep a full trajectory. Because consecutive plies of each game are analysed, the value of a played move lying outside the top- K is read off from the position it leads to—the next sampled decision—rather than re-searched separately: its win probability at depth d equals one minus the successor’s best win probability at depth $d-1$ (a half-move shallower, matching the other candidates’ negamax depth). On the rare occasions when no analysed successor exists (game end or the final sampled ply) the played move is assigned the weakest retained candidate’s win probability at each depth. With UCI_ShowWDL the engine reports win/draw/loss counts (w, dr, ℓ) in per-mille, so the win probability $P_{i,d} = (w + \frac{1}{2} dr)/1000$ is read off per move per depth—no logistic fit. Depth- d regret is

$$\delta_{i,d} = \max_j P_{j,d} - P_{i,d} \geq 0. \quad (1)$$

Because δ is a difference against the best move *at each depth*, it is invariant to anything that shifts all moves together (position difficulty, evaluation scale), isolating the choice problem. Searches run to depth D or a 2×10^8 -node cap, whichever is first, a safety bound on the rare quiet endgames whose multi-PV search does not converge; 0.09% of positions hit the cap and stop short of D , and are handled by masking the unreached depths in the likelihood (their reached depth is recorded) rather than imputing.

State-tower features. The human-pattern prior $q_i(p, c)$ is the Maia-3 policy at the side-to-move’s rating, read from the policy head as a full distribution over legal moves.

Latent-depth choice model. A player who has effectively searched to depth d chooses by a soft-max over depth- d regret fused with the pattern prior as a product of experts,

$$P(\text{move}=i \mid d, p, c) = \frac{\exp(-\beta \delta_{i,d} + \alpha \log q_i)}{\sum_j \exp(-\beta \delta_{j,d} + \alpha \log q_j)}. \quad (2)$$

Effective depth is latent with distribution $\pi_d(p, c) = P(\text{searched to } d \mid p, c)$, emitted by a network over a coarse depth grid conditioned on context c (rating, time class, clock, move number, total swing). The move likelihood marginalises depth,

$$P(\text{move}=i \mid p, c) = \sum_d \pi_d(p, c) P(\text{move}=i \mid d, p, c). \quad (3)$$

Parameters are fit by minimising the negative log-likelihood of the played move y , $\mathcal{L} = -\log \sum_d \pi_d P(y \mid d)$, with an entropy penalty on π for identifiability and β tied globally; data are split by player.

Depth of satisficing. The posterior over effective depth given the observed move,

$$r_d = P(d \mid y, p, c) = \frac{\pi_d P(y \mid d)}{\sum_{d'} \pi_{d'} P(y \mid d')}, \quad \hat{d} = \sum_d d r_d, \quad (4)$$

gives a per-decision depth with uncertainty. Depth-vs-rating is reported *within* time control (pooling confounds skill with available clock); per-band estimates use player-clustered bootstrap 95% credible intervals (resampling players, 1000 replicates).

Profiles and rating recovery. Per player (≥ 100 decisions): mean $\hat{d}$, trap-susceptibility (rate of played swing-down moves weighted by their shallow attractiveness), deep-discovery rate (rate of played swing-up moves), and time-elasticity (within-player slope of $\hat{d}$ vs. log think-time). To keep rating recovery non-circular, $\hat{d}$ for the profile is taken from a model fit with the rating feature removed. Rating is predicted from the standardised profile by ridge regression under nested 5-fold cross-validation (inner fold selects the penalty). The champion comparison reports the standardised components separately.

Model comparison and the differential test. State-only (Maia-3, $\beta=0$), search-only ($\alpha=0$), and fusion predictors are fit under one player split and compared on held-out cross-entropy and top-1 match. The pre-registered hypothesis is an interaction—the fusion’s gain over state-only is larger for high-swing than low-swing positions and for classical than blitz—tested with a player-clustered bootstrap and a linear mixed-effects model with players as random effects (positions are not crossed at this scale).

Convergent and recovery validation. Convergent: refit the model with the clock feature zeroed, then correlate per-decision $\hat{d}$ with observed time-on-move (held out, by game phase). Recovery: agents play a soft-max ($\beta_{\text{gen}}=6$) over fixed-depth d_{plant} win-probabilities; their games pass through the same pipeline, and a per-agent flat-prior posterior over depth (on the search likelihood, with the population sharpness β fit) is compared to d_{plant} .

Reproducibility. Stockfish 18 (net nn-71d6d32cb962.nnue), Threads 1, Hash 256 MB, $D=21$, $K=9$, 2×10^8 -node cap (positions analysed independently across many single-thread processes); Maia-3 maia3-79m; depth grid $\{2, 4, \dots, 22\}$; Lichess month 2025-09; sampling seed 17; train/test split by player. Strata and the primary interaction were fixed before held-out evaluation; we report results as confirmatory-on-held-out and label any post-hoc choices as exploratory.

Data and code availability. All game data derive from the public Lichess open database (<https://database.lichess.org>) and Lichess broadcast exports, which are openly licensed; the engine (Stockfish 18) and Maia-3 are open source. Analysis code, configuration, and figure scripts are released at [\[repository URL\]](#).

Ethics. The study uses only publicly released, openly licensed game records of voluntary online and broadcast play; it involves no intervention and no private or personally identifying data beyond the public usernames/player names attached to those records, and so did not require ethical approval.

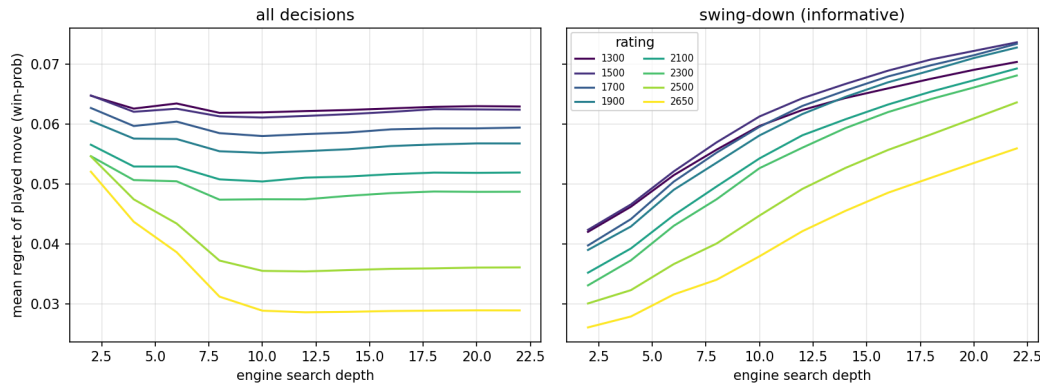


Figure 1: Played-move regret across engine search depth, by rating band, for all decisions (left) and swing-down decisions (right). Skill separates only at depth.

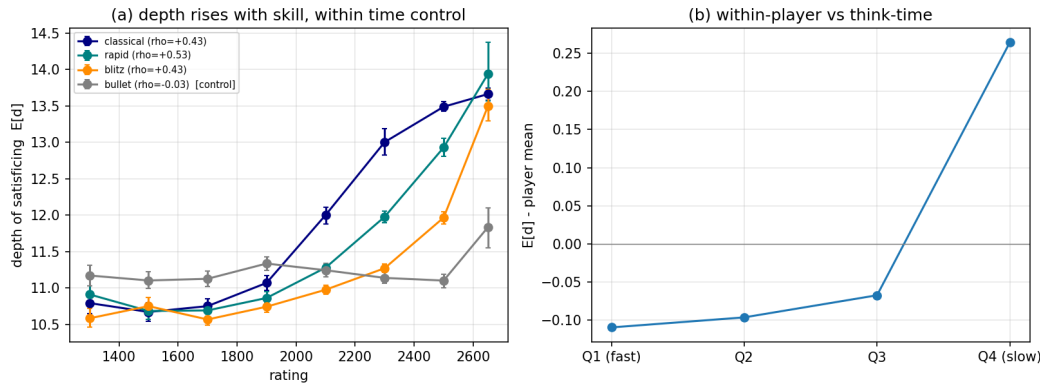


Figure 2: (a) Depth of satisficing vs. rating, within each time control (bullet is the flat negative control); player-clustered 95% CIs. (b) Within-player depth by think-time quartile.

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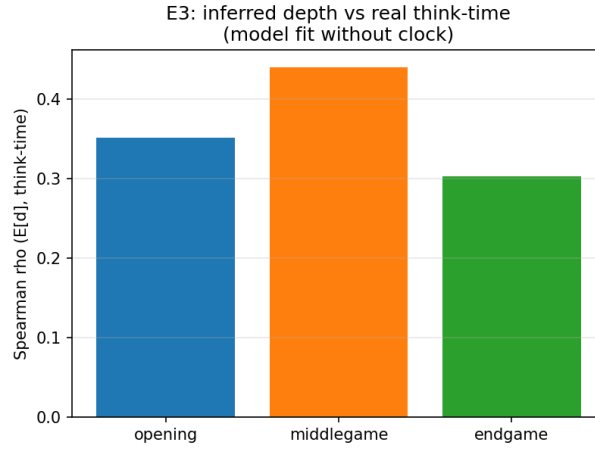


Figure 3: Convergent validity: clock-free inferred depth vs. observed time-on-move, by phase.

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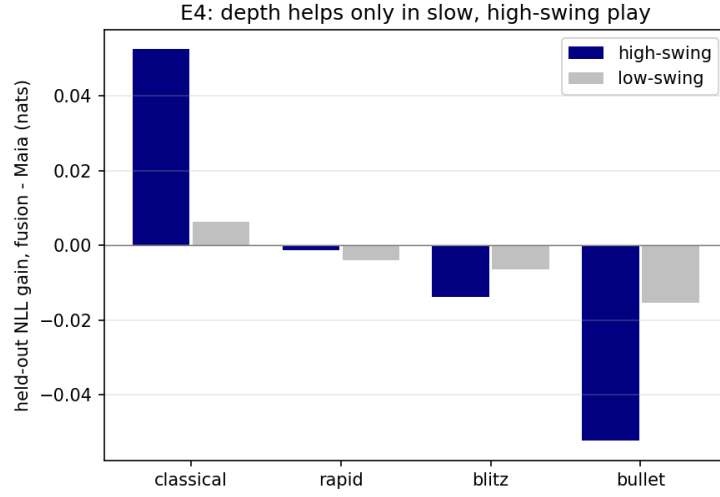


Figure 4: Held-out prediction gain of fusion over Maia-3, by time control $\times$ swing magnitude. Positive only in classical, high-swing play.

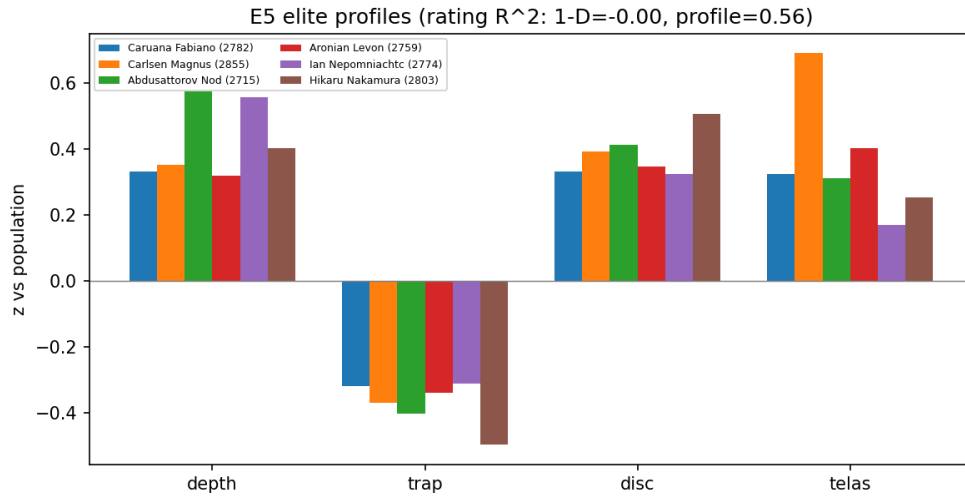


Figure 5: Elite four-axis profiles (z-scored); a one-dimensional depth score misranks them, the profile recovers rating ($R^2 = 0.57$).

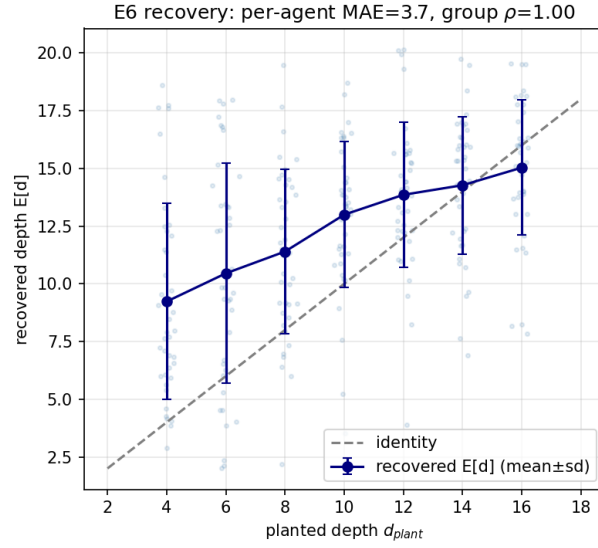


Figure 6: Synthetic-agent recovery: recovered vs. planted depth (ordinal; group Spearman = 1.0).

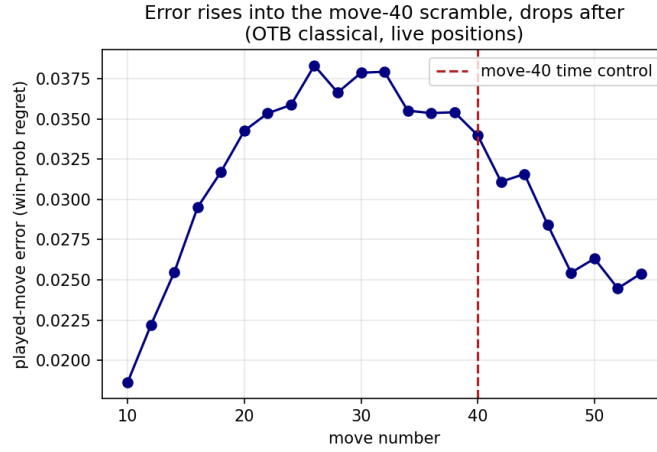


Figure 7: Within-game time-pressure signature: played-move error in live OTB positions rises into the move-40 control and drops once the clock is replenished.